

Principal Component Residual Trading on the U.S. Treasury Curve Signals, Hedging and Transaction Costs

Litterman and Scheinkman (1991) showed that around 96% of the variance in U.S. Treasury yield changes is explained by the first three principal components, economically interpreted as level, slope and curvature of the treasury yield curve. The residuals are the idiosyncratic deviations from the reconstructed treasury curve, if such deviations mean-revert within a tradable horizon the residuals form a tradable signal.

Data

We use daily constant maturity U.S. Treasury yields from the Federal Reserve (FRED-api) covering ten tenors from 3M to 30Y over 2006-2026. The yields are in bps and the backtest period begins in 2016 to allow for a sufficient burn-in window for the correlation matrix and the eigen decomposition.

Principal Components and Residuals

For each day, we fit a 5-year rolling correlation matrix R on standardized Treasury yield levels, storing the tenors' mean and standard deviation used for standardization. Because R is symmetric, its eigen decomposition yields an orthonormal eigenvector matrix Q :

$$R = Q\Lambda Q^T$$
$$V = \begin{bmatrix} \vdots & \vdots & \vdots \\ Q_1 & Q_2 & Q_3 \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad V \in \mathbb{R}^{10 \times 3}$$

We only retain the matrix V , containing the largest three eigen vectors of the matrix Q . Furthermore to ensure consistent factor interpretation across rolling windows, we sign-aligned the eigenvectors: V_1 is oriented such that the 10Y loading is positive, V_2 such that the 30Y loading is bigger than the 2Y loading, and V_3 such that the 5Y loading is negative.

Let X be the daily standardized yield vector:

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_{10} \end{bmatrix}, \quad \tilde{X} = \begin{bmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_{10} \end{bmatrix}, \quad \tilde{x}_i = \frac{x_i - \mu_i}{\sigma_i}$$

Projecting $\tilde{X}$ onto V results in $\hat{\tilde{X}}$. Where $\hat{\tilde{X}}$ is the standardized yield vector with components that can be explained by the first three principal components: level, slope and curvature. Because the columns of V are orthonormal, this projection reduces to

$$\hat{\tilde{X}} = VV^T \tilde{X}$$

Destandardizing $\tilde{X}$ results in $\hat{X}$:

$$\hat{X} = \begin{bmatrix} \hat{x}_1 \\ \vdots \\ \hat{x}_{10} \end{bmatrix}, \quad \hat{x}_i = \tilde{x}_i \sigma_i + \mu_i$$

Now we can obtain the daily residuals vector ε :

$$\varepsilon = X - \hat{X}$$

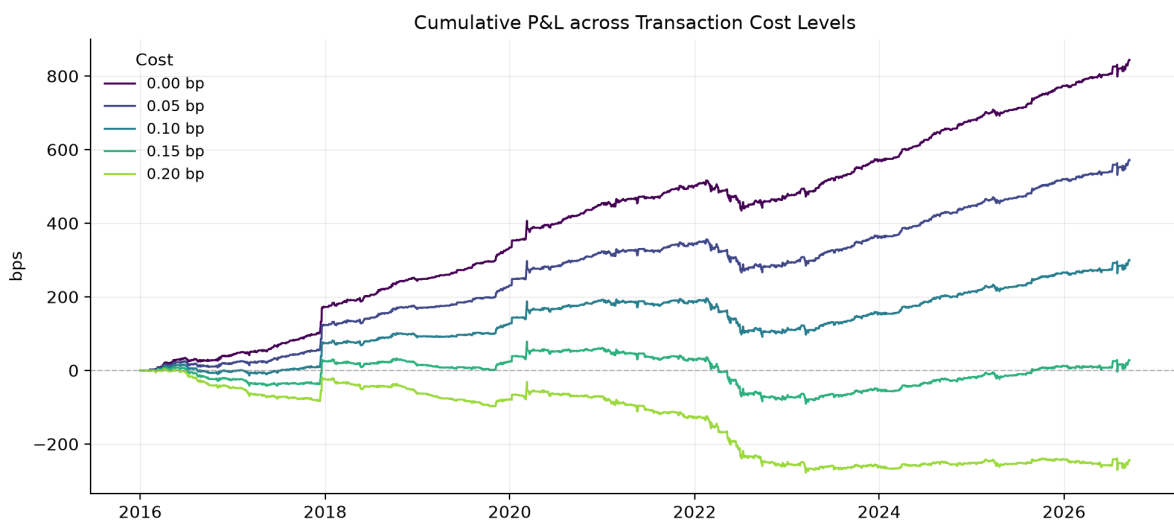
Signal Generation and Hedging

We used the residual series for each tenor and calculated the rolling z-score over a 20-day period, where a z-score of +2 gives the signal to short the tenor's yield and -2 to long the yield. The positions were closed when the z-score reached +0.5 for a short position and -0.5 for a long position. Shorting yields corresponds to being long on a bond and vice versa, due to the inverse relationship of bond prices and yields. We only used the tenors from 2Y until 20Y to ensure we have neighboring tenors to use for the hedge and avoid the lower tenors where price movement is dominated by Fed-policy.

Since we only want to trade the deviations without being exposed to the three principal components, we hedge the traded tenor's principal component with three surrounding tenors. We know $V \in \mathbb{R}^{10 \times 3}$, where entry $V_{i,j}$ is the loading of tenor i on eigenvector j . let $W \in \mathbb{R}^{3 \times 3}$ denote the Transpose of V with only the rows of the tenors we use for the hedge. And let $P \in \mathbb{R}^3$ denote the transpose of the row vector of V corresponding to the tenor we are trading and thus trying to hedge. We solve the following equation for $H \in \mathbb{R}^3$, which denotes the vector containing the position we must take in the tenors used for hedging to be principal component neutral.

$$WH = -P$$

Results and Transaction Costs Sensitivity



cost (bps/leg)	cumulative P&L (bps)	modified Sharpe
0.0	843.41	2.01
0.05	571.41	1.37
0.1	299.41	0.72
0.15	27.41	0.07
0.2	-244.59	-0.59

$$\text{modified sharpe ratio} = \frac{\text{mean daily return in bps}}{\text{standard deviation of daily returns in bps}} \sqrt{252}$$

We call this a modified Sharpe ratio, as it does not include a risk-free rate. A conventional Sharpe compares excess return over the risk-free rate divided by volatility, which assumes the capital could alternatively earn the risk-free rate. This strategy is not a long-only portfolio: it consists of offsetting long and short positions, which causes partly self-financed positions. So for the scope of this metric it suffices to assume a fully self-financed position.

Furthermore, basis points are treated equally across tenors in the P&L, whereas in monetary terms a single-bp move corresponds to different dollar gains depending on the tenor's duration.

Interpretation and Limitations

This strategy breaks even at around 0.15 bp per leg and is only truly viable below 0.10 bp. Such execution costs are realistic for a select group of market participants, like HFT firms, not for the broader cross-section of market participants, for whom it is a losing strategy. This is consistent with the size and efficiency of the U.S. Treasury market.

Even setting transaction costs aside, there still are several caveats. First, the yield data are not point-in-time and may include revisions, which can introduce a forward-looking bias. Second, the P&L is expressed in basis points, which implicitly weights all tenors equally in dollar terms. Third, the analysis fixes a single parameter set (5-year PCA window, 20-day z-score window, $\pm 2/\pm 0.5$ entry-exit thresholds), a broader robustness analysis across these parameters would be required to separate real signal from a favorable sample.